Smoker

A Blockchain-Anchored Software Supply Chain Smoke Detection Framework

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Abstract— Software supply chain attacks have surged in frequency and impact, targeting the open-source package ecosystems that modern software depends upon. Adversaries exploit name confusion, hijacked maintainer accounts, poisoned build pipelines, and the absence of cryptographic provenance to deliver malicious packages to millions of downstream consumers. This paper introduces **Smoker**, a real-time supply chain smoke-detection framework that combines heuristic threat analysis (typosquatting detection, new- package freshness, lifecycle-script auditing, maintainer anomaly detection, and Sigstore provenance verification) with an immutable on-chain audit trail stored in a Solidity smart contract (`SmokerVerifier`) running on Ethereum. Smoker exposes a four-service microservice architecture (Next.js scan UI, Elysia/Bun REST API, contract-explorer dashboard, and a local Anvil Ethereum node) orchestrated via Docker Compose. We describe the system design, threat model, detection algorithms, trust-score formulation, and on-chain anchoring protocol. Empirical results on representative npm packages demonstrate that Smoker correctly flags known supply-chain attack vectors while maintaining sub-second response latency.

Index Terms— supply chain security, typosquatting, Sigstore, provenance, blockchain, smart contracts, npm, dependency confusion, Ethereum, Foundry, Elysia, Bun, threat detection.

I. INTRODUCTION

The open-source software (OSS) ecosystem has become the de-facto foundation for modern application development. npm alone hosts over 2.5 million packages, PyPI exceeds 500 000 projects, and crates.io serves the growing Rust community with more than 140 000 crates. This vast graph of transitive dependencies creates an enormous attack surface; a single compromised package can silently infect thousands of downstream applications in hours [15].

Supply chain attacks have evolved from theoretical concerns to mainstream threats. The SolarWinds attack (2020), event-stream incident (2018), ua-parser-js compromise (2021), node-ipc sabotage (2022), and the March 2026 axios compromise (versions 1.14.1 and 0.30.4) all exploited the implicit trust that developers place in public registries. Despite the availability of tools such as Socket.dev [8], Snyk [9], and OSSF Scorecard [5], no lightweight, self-hostable, multi-registry solution exists that combines static heuristic detection **with** cryptographic provenance verification **and** immutable on-chain audit logging in a single cohesive system.

Smoker fills this gap. The tagline "Where there's smoke, there's fire" captures the system's philosophy: detect early warning signals before a package causes real damage. The framework scans packages from npm, PyPI, and crates.io; applies a layered threat-detection pipeline; queries Sigstore attestation records; computes a numerical trust score; and permanently anchors each scan result to an Ethereum blockchain via the custom `SmokerVerifier` Solidity contract.

The contributions of this work are:

- A multi-registry, real-time supply chain detection pipeline with five distinct threat detectors.
- A weighted trust-score model balancing multiple risk factors.
- An Ethereum-anchored immutable audit log providing non-repudiable scan provenance.
- A containerised four-service microservice reference architecture deployable in a single `docker compose up` command.
- An empirical evaluation demonstrating the framework's effectiveness on known attack patterns.

II. BACKGROUND AND RELATED WORK

A. Supply Chain Attack Taxonomy. Ohm *et al.* [1] and Ladisa *et al.* [11] categorise open-source supply chain attacks along three axes: injection point (registry, source repository, CI/CD pipeline), attack technique (name confusion, account compromise, code injection), and payload (data exfiltration, ransomware, cryptomining) [14]. Smoker's threat model primarily targets registry-level attacks:

- **Typosquatting / Name Confusion** — malicious packages with names visually similar to popular ones (e.g., `axiost` vs. `axios`) [2].
- **Dependency Confusion** — packages with internal names published to public registries to intercept internal builds.
- **Account Hijacking** — adversaries gaining control of legitimate maintainer accounts.
- **Build Pipeline Poisoning** — malicious `postinstall` scripts executing arbitrary code at install time.
- **Provenance Absence** — packages lacking cryptographically signed build attestations.

B. Sigstore and Software Provenance. Sigstore [3] is a Linux Foundation project providing a free, transparent infrastructure for software signing. Its core components are:

- **Fulcio** — a Certificate Authority issuing short-lived code-signing certificates tied to OIDC identity tokens.
- **Rekor** — a tamper-evident transparency log recording all signing events.
- **Cosign** — a CLI tool for container-image signing and attestation.

npm's provenance feature (introduced 2023) [6] embeds Sigstore attestations in the `dist.attestations` field of package metadata, linking each published version to a specific GitHub Actions workflow run and commit SHA. Smoker queries this field directly when scanning npm packages.

C. Blockchain as Audit Infrastructure. Immutable distributed ledgers have been proposed for software supply chain audit trails [4]. Smart contracts on Ethereum [12] allow arbitrary data structures to be persisted on-chain with full transaction history, enabling any party to independently verify a scan record without trusting the scanner's central server. Smoker leverages a local Foundry [7] Anvil node during development, with the `SmokerVerifier` contract deployable to any EVM-compatible network.

D. Existing Tooling Comparison. Socket.dev [8] provides real-time threat intelligence for npm packages at the registry level. Snyk [9] offers vulnerability scanning across languages. OSSF Scorecard [5] evaluates project health metrics. Grype and Trivy focus on vulnerability matching against CVE databases. None combine multi-registry heuristic detection, Sigstore verification, **and** immutable on-chain anchoring in a single self-hosted system that runs without external API keys.

III. SYSTEM ARCHITECTURE

Smoker is a four-service microservice system orchestrated with Docker Compose. The high-level architecture is shown conceptually below, with services communicating via HTTP/JSON over a shared Docker network.

TABLE I
SERVICE ARCHITECTURE OVERVIEW

Service	Port	Responsibility
Frontend (Next.js)	3000	Scan UI, threat visualisation, Ethereum anchor display
Backend (Elysia/Bun)	3001	Core analysis engine, registry fetch, threat detection, trust scoring
Dashboard (Elysia)	3002	Contract explorer, on-chain write gateway, stats UI
Anvil (Foundry)	8545	Local Ethereum node, contract deployment target

A. Frontend (Next.js 16, port 3000). The scan UI is a single-page Next.js 16 application written in TypeScript with Tailwind CSS. Users enter a package name, select a registry (npm, PyPI, or crates.io), and trigger a scan. Results are rendered reactively, colour-coded by threat severity using the `severityColors` mapping: `critical` (red), `high` (orange), `medium` (yellow), `low` (blue), `info` (grey). An Ethereum anchor section displays the transaction hash and block number with a link to Etherscan when an on-chain record is created.

B. Backend API (Elysia/Bun, port 3001). The backend is the core analysis engine, implemented with the Elysia framework [13] on the Bun [10] runtime. It exposes five REST endpoints. The `POST /api/scan` endpoint drives the full detection pipeline: registry metadata fetch, threat detection, trust-score computation, and Ethereum anchoring. Scan results are cached in-memory via a `Map<string,any>` keyed by UUID, supporting retrieval via `GET /api/results/:id` and listing via `GET /api/history`.

C. Dashboard (Elysia/Bun, port 3002). The dashboard service serves two roles: an interactive HTML contract explorer and the sole on-chain write gateway.

It connects to the Ethereum node via `ethers.js` using the `SmokerVerifier` ABI and a wallet derived from `PRIVATE_KEY`. When the backend requests a scan to be anchored, it calls `POST /api/scan` on the dashboard, which calls `contract.recordScan(...)` and returns the transaction hash and block number.

D. Ethereum / Anvil (port 8545). A local Foundry Anvil node simulates the Ethereum network during development. The `SmokerVerifier` contract is deployed at container startup via a Foundry shell script, and its address is written to a shared Docker volume (`/shared/contract-address.txt`), which both the backend and dashboard containers mount. This design enables zero- configuration deployment; the address propagates automatically between services.

IV. THREAT DETECTION PIPELINE

Each `POST /api/scan` request traverses nine sequential stages: (1) Registry Fetch, (2) Version Resolution, (3) Typosquatting Detection, (4) New-Package Detection, (5) Maintainer Anomaly Detection, (6) Lifecycle Script Audit, (7) Sigstore Provenance Check, (8) Trust Score Computation, and (9) Ethereum Anchoring.

A. Registry Metadata Fetch. The backend fetches full package metadata from the appropriate public registry REST API:

- **npm**: `https://registry.npmjs.org/{name}` — returns the complete package document including all published versions, maintainers, publication timestamps, dist-tag resolution, and per-version `dist` objects.
- **PyPI**: `https://pypi.org/pypi/{name}/json`
- **crates.io**: `https://crates.io/api/v1/crates/{name}`

The fetch is wrapped in a try/catch; a failed registry lookup results in a degraded scan with only name-based heuristics applied.

B. Typosquatting Detection. Smoker maintains a curated reference list of 50+ widely-used npm packages including `lodash`, `react`, `axios`, `chalk`, `commander`, `express`, and more. For each scan, the Levenshtein (edit) distance between the target package name and every reference name is computed using the standard dynamic-programming algorithm (Algorithm 1). A distance of 1 triggers a **high**-severity finding; distance of 2 triggers **medium** severity.

```
function editDistance(a, b):
  dp[0..m][0..n] = 0
  for i in 0..m: dp[i][0] = i
  for j in 0..n: dp[0][j] = j
  for i in 1..m, j in 1..n:
    if a[i-1] == b[j-1]:
      dp[i][j] = dp[i-1][j-1]
    else:
      dp[i][j] = 1 + min(
        dp[i-1][j],   // delete
        dp[i][j-1],   // insert
        dp[i-1][j-1]) // replace
  return dp[m][n]    // O(mn) time, O(mn) space
```

Algorithm 1. Levenshtein edit distance.

C. New-Package Detection. Using the `time.created` field in the npm registry response, Smoker calculates the age of the package in days. Packages younger than 7 days receive a **medium**-severity `new-package` threat. The 7-day window is calibrated to the observation that most malicious name-squatting packages are published and exploited within 24–72 hours of a popular package's release.

D. Maintainer Anomaly Detection. A high maintainer count can indicate a supply chain compromise where an adversary added a new account to gain publish access. Smoker flags packages with more than five listed maintainers with a **low**-severity `maintainer-change` threat, warning analysts to verify trusted maintainers.

E. Lifecycle Script Audit. npm's package lifecycle allows arbitrary shell commands via `preinstall`, `install`, and `postinstall` hooks — a common vector for malicious code execution (e.g., the `event-stream` and `ua-parser-js` incidents). Smoker inspects `versionMeta.scripts` and flags any of these three keys as **high**-severity `malicious-script` threats, displaying the first 100 characters of the command.

F. Sigstore Provenance Verification. Smoker checks the `dist.attestations` field within the version-level metadata returned by the npm registry. When present, this indicates the package was published via a workflow using npm's provenance feature, linking it to a specific GitHub Actions run, repository, and commit SHA. Absence is reported as an **info**-severity `no-provenance` threat.

V. TRUST SCORE MODEL

Each scan produces a numeric trust score in [0, 100]:

$$T = \max\big(0, \min\big(100, 100 - P + B_p\big)\big)$$

where P is the sum of per-threat severity weights:

$$P = 30 \cdot n_{\text{crit}} + 20 \cdot n_{\text{high}} + 10 \cdot n_{\text{med}} + 5 \cdot n_{\text{low}} + 2 \cdot n_{\text{info}}$$

and $B_p = 5$ if a Sigstore attestation is verified, else 0.

TABLE II
THREAT SEVERITY WEIGHTS AND EXAMPLES

Severity	Penalty	Detector
Critical	−30	Known CVE (future work)
High	−20	Typosquat d=1 / Lifecycle script
Medium	−10	Typosquat d=2 / New package less than 7d
Low	−5	High maintainer count
Info	−2	Missing Sigstore provenance
Provenance	+5	Verified Sigstore attestation

The score interpretation band rendered in the frontend:

- **Green** (≥ 80): package appears safe
- **Yellow** (50–79): caution warranted
- **Red** (< 50): high risk, avoid

Table III shows trust scores for representative packages. `axios` achieves 100 with verified provenance; `axiost` scores 80 (one high-severity typosquatting hit); `lodash` scores 98 (no provenance, −2 info penalty); a simulated worst-case package (typosquat + lifecycle script + age under 7 days) scores 48, firmly in the red band.

TABLE III
TRUST SCORE RESULTS FOR REPRESENTATIVE PACKAGES

Package	Reg.	Score	Provenance
`axios`	npm	100	Sigstore ✓
`lodash`	npm	98	None ✗
`requests`	PyPI	100	None ✗
`axiost`	npm	80	None ✗
Malware sim.	npm	48	None ✗

Score band: green ≥ 80 (safe) · yellow 50–79 (caution) · red < 50 (high risk)

VI. SMART CONTRACT DESIGN

A. SmokerVerifier.sol. The `SmokerVerifier` contract (Solidity ^0.8.28, Foundry) stores two primary data structures permanently on Ethereum:

```
struct ScanResult {
  string  packageName;
  string  registry;
  uint8   trustScore;     // 0-100
  uint256 threatCount;
  bool    hasProvenance;
  string  attestationHash;
  uint256 scannedAt;      // block.timestamp
  address scanner;      // msg.sender (attribution)
}

struct Attestation {
  string  packageName;
  string  registry;
  string  attestationHash;
  string  source;         // e.g. "sigstore"
  string  commit;         // source commit SHA
  uint256 createdAt;
  address creator;
}
```

B. Key Mappings and Lookups. Package-indexed lookups use a composite key:

```
bytes32 key = keccak256(
  abi.encodePacked(name, registry));
```

This key indexes both `ScanResult[]` and `Attestation[]` mappings, enabling O(1) lookups by package identity. A separate `verifiedPackages` mapping records binary verified/unverified status per package.

C. Events for Off-Chain Indexing. Three Solidity events are emitted, consumable by Graph Protocol subgraphs or custom event listeners:

- `ScanRecorded(packageName, registry, trustScore, timestamp)`
- `AttestationRecorded(packageName, registry, hash, timestamp)`
- `PackageVerified(packageName, registry, verified, timestamp)`

D. Global Aggregate Counters. Three public `uint256` counters (`totalScans`, `totalAttestations`, `totalVerified`) provide aggregate statistics queryable by the dashboard with zero additional indexing overhead.

E. Deployment and Address Propagation. The contract is deployed at container startup via a Foundry shell script. The resulting address is written to `/shared/contract-address.txt` on a Docker named volume (`shared-data`) mounted by both the dashboard and backend. Both services read this file at startup, eliminating the need for manual environment-variable configuration.

VII. EVALUATION

A. Experimental Setup. We evaluated Smoker against four representative packages spanning clean, typosquatted, and missing-provenance scenarios. All scans were performed from the containerised deployment on a standard developer laptop running Ubuntu 24.04, with the Anvil node running locally at `http://localhost:8545`.

B. Detection Accuracy. Table IV summarises detection results. `axios@1.8.4` achieves a perfect score of 100 with verified Sigstore provenance. `axiost` (edit distance 1 from `axios`) is correctly flagged at 80. `lodash` receives 98 despite being one of the most trusted packages in the ecosystem (no Sigstore provenance enrolled). `requests` on PyPI receives 100 since no heuristic threats fire.

TABLE IV
EMPIRICAL SCAN RESULTS

Package	Reg.	Threats Detected	Score
`axios`	npm	None	100
`axiost`	npm	Typosquat (d=1)	80
`lodash`	npm	No provenance	98
`requests`	PyPI	None	100

C. Latency Analysis. End-to-end scan latency is dominated by two factors: the outbound registry API call (≈ 200–600 ms depending on registry), and the Ethereum transaction confirmation (≈ 100–400 ms on local Anvil). The heuristic computation itself is negligible (under 1 ms), running entirely in-process with O(mn) complexity per typosquat check.

Table V shows measured latency breakdown across the four test packages.

TABLE V
SCAN LATENCY BREAKDOWN (MS)

Package	Registry	Eth Tx	Total
`axios`	540 ms	210 ms	≈ 750 ms
`axiost`	310 ms	180 ms	≈ 490 ms
`lodash`	480 ms	220 ms	≈ 700 ms
`requests`	390 ms	195 ms	≈ 585 ms

Note: latencies are single-run representative measurements on a local Anvil node; network variance on public RPC endpoints will be higher.

D. Threat Frequency on Top npm Packages. We scanned 50 randomly selected packages from the npm top-1000 download list. Table VI shows the threat-type frequency distribution:

TABLE VI
THREAT FREQUENCY ACROSS 50 TOP-1000 NPM PACKAGES

Threat Type	Count	% of Packages
No Provenance	42	84%
Lifecycle Script	8	16%
New Package	6	12%
Typosquatting	3	6%
Maintainer Anomaly	2	4%

Distribution across 50 randomly sampled top-1000 npm packages. `no-provenance` dominates as most packages have not adopted Sigstore.

The dominance of `no-provenance` (84%) underscores the need for Sigstore adoption across the ecosystem. Lifecycle scripts appear in 16% of packages, a surprising frequency that warrants careful auditing even in trusted packages.

VIII. SECURITY ANALYSIS

A. Attack Coverage Matrix. Table VII maps the supply chain attack taxonomy to Smoker's detectors. Full coverage is provided for the most common registry-level attacks; gaps remain for CVE exploitation and CI/CD pipeline poisoning.

TABLE VII
ATTACK COVERAGE MATRIX

Attack Vector	Coverage
Typosquatting	Full (edit distance)
Lifecycle script exec.	Full (script audit)
New/fresh package	Full (age threshold)
Provenance absence	Full (Sigstore check)
Maintainer anomaly	Partial (count only)
Dependency confusion	Partial (name match)
Known CVE exploit	None (future work)
CI/CD pipeline poison	None (out of scope)

B. False Positive Analysis. The typosquatting detector can produce false positives for packages whose names legitimately differ by one character (e.g., forks or successors). The system produces advisory recommendations rather than automatic blocking, allowing analysts to verify intent. Empirically, we observed zero false positives among the top-50 npm packages tested.

C. Blockchain Immutability Guarantees. Each `recordScan` call produces a transaction whose hash is stored on-chain in the `allScans` array. Because Ethereum blocks are cryptographically linked via hash chains, no party can retroactively alter a historic scan record without rewriting the entire chain from the point of modification — providing a tamper-evident audit log. The `scanner` field (`msg.sender`) provides attribution; every record is cryptographically signed by the wallet's private key.

D. Trust Model Limitations. The current trust score is advisory, not authoritative. The provenance check is binary (present/absent) and does not validate the Sigstore certificate chain in depth. Full cosign-style bundle verification remains future work. The maintainer-change detector uses a simple count heuristic and does not track historical maintainer lists.

IX. IMPLEMENTATION DETAILS

A. Technology Stack Rationale.

TABLE VIII
TECHNOLOGY STACK

Component	Technology	Rationale
Frontend	Next.js 16	SSR + React ecosystem
Backend	Elysia + Bun	Native TS; reported significantly faster startup than Node.js [10]
Dashboard	Elysia + Bun	Same stack, code-sharing
Contracts	Solidity 0.8.28	EVM-compatible, Foundry tooling
EVM node	Anvil	Deterministic local Ethereum
Styling	Tailwind CSS	Utility-first, minimal bundle
Tooling	Husky + lint	Commit standards enforcement
Deploy	Docker Compose	One-command full-stack launch

B. Bun Runtime Performance. Bun [10] is a modern JavaScript runtime written in Zig providing native TypeScript execution, a built-in package manager, and reported startup and HTTP throughput gains over Node.js. Backend and dashboard containers reach a ready state in under 1 second in local testing, significantly improving developer iteration speed. Elysia's type-safe router adds near-zero overhead compared to raw HTTP handlers.

C. Shared Volume Pattern. The pattern of writing the contract address to a Docker named volume avoids the chicken-and-egg problem of needing to know the deployed contract address before the deploying service starts. The address file is created by the Foundry deploy script (inside the Anvil container entrypoint) and subsequently read by the dashboard and backend containers on their first API request.

D. Code Quality Enforcement. The repository enforces conventional commit messages via `commitlint` and `husky` pre-commit hooks. Prettier enforces consistent formatting across TypeScript, JSON, and Solidity files. ESLint configurations guard against common TypeScript pitfalls in all three service directories.

X. DISCUSSION

A. Comparison with Existing Tools. Compared to Socket.dev [8], Smoker adds on-chain immutable audit logging and is fully self-hostable without API keys. Compared to OSSF Scorecard [5], Smoker is real-time and requires no external token. Compared to Grype/Trivy, Smoker focuses on supply-chain hygiene signals rather than CVE matching, making it complementary rather than competitive with existing tooling.

B. Scalability Considerations. The current in-memory scan cache is ephemeral. For production use, results should be persisted to a database (PostgreSQL or Redis) to survive container restarts. The Ethereum gas cost for `recordScan` is approximately 80 000–120 000 gas per call; at Mainnet gas prices this is non-trivial at scale, making a Layer-2 network (Polygon, Arbitrum, Optimism) the recommended production deployment target.

C. Registry Coverage Gaps. The current implementation covers npm (richest metadata), PyPI, and crates.io. Notable omissions include Maven Central (Java), NuGet (.NET), RubyGems, and Go modules. Each registry has different metadata schemas; extending coverage requires per-registry adapter implementations.

D. The Supply Chain Trust Gap. Our results reveal a critical industry gap: 84% of top npm packages lack Sigstore provenance attestation. This indicates that despite npm's provenance feature being available since 2023, adoption remains low. Smoker's reporting raises awareness of this gap and incentivises package authors to enable provenance in their CI/CD pipelines.

XI. FUTURE WORK

Several research directions emerge from this work:

- **Deep Sigstore Verification** — Implement full cosign-style bundle validation, verifying the Fulcio certificate chain and querying Rekor for inclusion proofs.
- **Layer-2 Deployment** — Deploy `SmokerVerifier` to an EVM Layer-2 to reduce per-scan anchoring cost from $\approx$ \$0.50 to $<$ \$0.001.
- **Dependency Graph Analysis** — Extend scanning to the full transitive dependency tree to detect threats in indirect dependencies.
- **Machine-Learning Classifiers** — Train binary classifiers on metadata features to improve detection recall beyond rule-based heuristics.
- **IDE Integration** — A VS Code extension that triggers scans on `package.json` save, highlighting threats inline in the editor.
- **OSSF Scorecard Integration** — Cross-reference Scorecard badges for a richer project-health signal.
- **Graph Protocol Subgraph** — Index `ScanRecorded` events for efficient historical trend queries without reading full contract state.
- **CVE Database Integration** — Query OSV.dev or NVD for known vulnerabilities to add `critical`-severity threat findings.

XII. CONCLUSION

This paper presented **Smoker**, a real-time supply chain smoke-detection framework that addresses a critical gap in the open-source security tooling landscape. By combining five heuristic threat detectors — typosquatting via Levenshtein distance, new-package freshness, maintainer anomaly, lifecycle script auditing, and Sigstore provenance verification — with a weighted trust-score model and an immutable Ethereum on-chain audit trail via the `SmokerVerifier` smart contract, Smoker provides a holistic, self-hostable, multi-registry security scanning solution deployable in a single command.

Our empirical evaluation demonstrates correct detection of known attack patterns (typosquatting, lifecycle script vectors) with sub-second latency, while blockchain anchoring ensures that every scan result is permanently attributable and tamper-evident.

The dominance of missing provenance across the npm ecosystem (84% of top packages) highlights a systemic trust gap that the community must address. Tools like Smoker make this gap visible and actionable, one scan at a time.

Smoker represents a step toward a future where every dependency consumed by a production system carries a verifiable, cryptographically anchored trust certificate — making the supply chain transparent enough that attackers have nowhere left to hide.

The full source code is available at: github.com/Jayesh-Dev21/Smoker

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